

Project Exoplanets

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Abstract

This project presents an analysis of exoplanet transit data from the Kepler mission in order to study the different characteristics of the exoplanet Kepler-10 b. The data are processed using Python and Jupyter Notebook to estimate planetary parameters.

Introduction

In practice, light curves representing the brightness of stars are analyzed. When an exoplanet passes in front of its star, it blocks a small fraction of the stellar light, causing a decrease in the observed flux. This phenomenon is known as a transit. The presence of periodic and regular dips in a light curve is therefore an indicator of an exoplanet transiting its star.^[1]

Literature Review

According to previous studies, Kepler-10 b is a very small planet compared to its star, Kepler-10. Kepler-10 measures 106 solar radii, or 73776000000 meters while Kepler-10 b measures 1.469 Earth radii, or 9380040 meters. This exoplanet has a very short orbital period, meaning it takes very little time to orbit its star Kepler-10 : it makes one orbit in 0.837491 days, or approximately 20 hours.^[2]

Several results are expected from the experiment. First, it should give a low transit depth corresponding to the fraction of light blocked by the

exoplanet, which is insignificant because of the difference in size between Kepler-10 b and its star. In addition, this experiment should give a radius for the exoplanet that is slightly larger than Earth's but insignificant compared to the radius of the star. Finally, the orbital distance obtained should be short, because the period found in the experiment must itself be very short, since these two data points are related to the proximity of Kepler-10 b and its star, as proven by the period of revolution of Kepler-10 b given above.

Method

All data analyses were performed using Python software (version 3.x) in a Jupyter Notebook environment. The NumPy and pandas libraries were used for data processing and data manipulation, Matplotlib for graph visualization, and the Lightcurve package for accessing and processing data from the Kepler mission.^[3]

The raw Kepler-10 flux data were obtained from NASA's Exoplanet Archive, based on observations made by the Kepler space telescope. The raw stellar flux was first plotted graphically over the entire duration of the Kepler observation campaign. The raw flow is sampled to simplify calculations and corrected by centering the flow around 1, thereby removing abnormal values. Folding and binning techniques make light curve transits visible. The folding method consists of defining a starting point (the epoch) that corresponds to the very first point on the curve (more generally, the beginning of a cycle). From there, the curve will be folded a certain number of times in order to superimpose all the cycles (for example, all the transits). The binning method consists of grouping data points into "small groups" and calculating the average in each group in order to smooth the curve.

Transit detection was performed using the BLS (Box Least Squares) algorithm.^{[4][5]} This method searches for periodic box-shaped decreases in the flux by analyzing a series of periods. For each period, the light curve is folded back and the depth of the transit is measured as the difference between the flux during transit and outside transit. The period corresponding to the largest transit signal is identified as the orbital period of the planet. Since the exact transit position is unknown, the box is systematically shifted along the folded light curve to measure the transit depth at each phase.

The BLS program is tested for the planet Kepler-10 b, and the obtained period was refined by scanning a narrow range of trial periods. It is used

to plot the light curve of Kepler-10 with a visible transit of Kepler-10 b. The transit depth is calculated using the same folded and binning data as the Kepler-10 light curve plot. The radius of this star is obtained using a formula containing the transit depth and orbital distance using Kepler's third law.

Results

The orbital period of Kepler-10 b obtained in this project after two successive scans of the tested periods was 0.8373737373737374 days (Figure 1), approximately 20,097 hours.

Estimated period : 0.8373737373737374 days
Maximum depth : 0.00019376955717675948

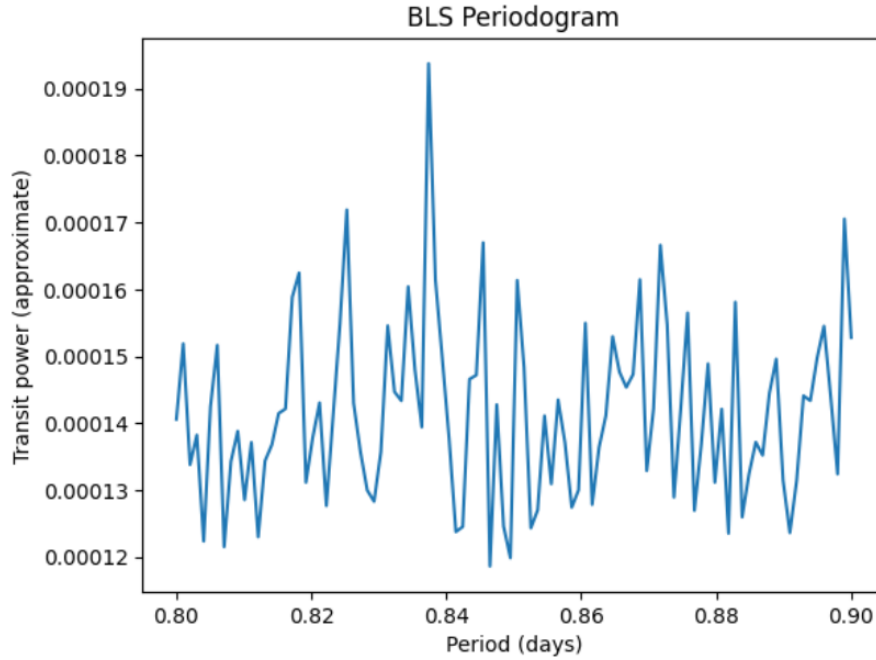


Figure 1: BLS Periodogram

The normalized transit depth obtained was $9.623623663924286 \cdot 10^{-05}$, approximately 0,0096 %. The radius of Kepler-10 b obtained was 7612565.404 meters and its orbital distance was 2506710662.043048 meters, or approximately 0.01675608731312198 AU.

Discussion/Analysis

The orbital period of Kepler-10 b is only 20.097 hours (which is almost identical to the 20 hours given by NASA). [6] This is very fast compared to Earth and its orbital period of 365 days around the Sun. Such a short period indicates a small orbital distance and therefore more frequent transits, making Kepler-10 b easier to detect. However, the BLS program of the experiment gives a much less pronounced transit than that of NASA. (Figures 2 and 3)

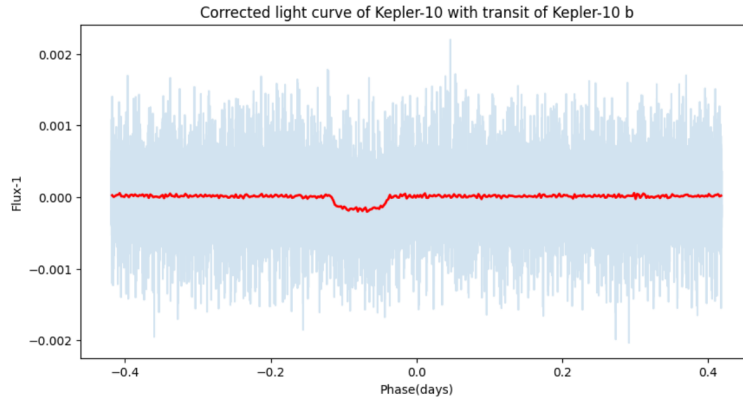


Figure 2: Corrected light curve of Kepler-10 with transit of Kepler-10 b

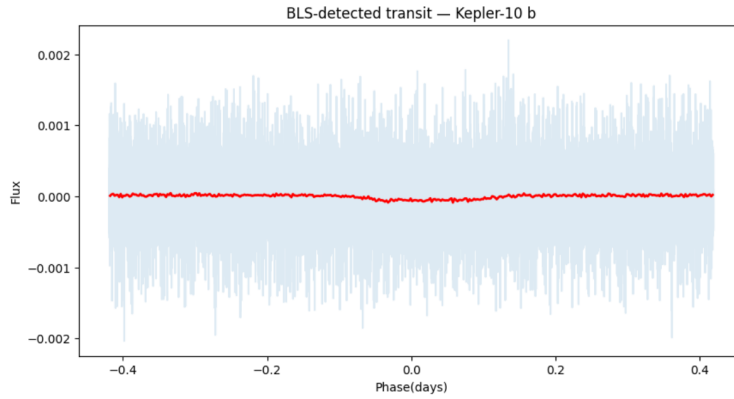


Figure 3: Folded light curve showing the transit of Kepler-10 b detected by the BLS

The orbital distance given by the experiment, very close to NASA's

0.01685 AU, confirms these hypotheses.^[6] The normalized transit depth is on the order of 10^{-5} , which means that Kepler-10 b blocks a very small fraction of its star's light. This depth value is broadly consistent with that of NASA (0.0093%).^[6] The slight differences between the depth values from the experiment and NASA are the result of a flux limited by data sampling, binning, folding, and instrumental noise, i.e., unwanted signals that interfere with the measurements. (Figure 4)

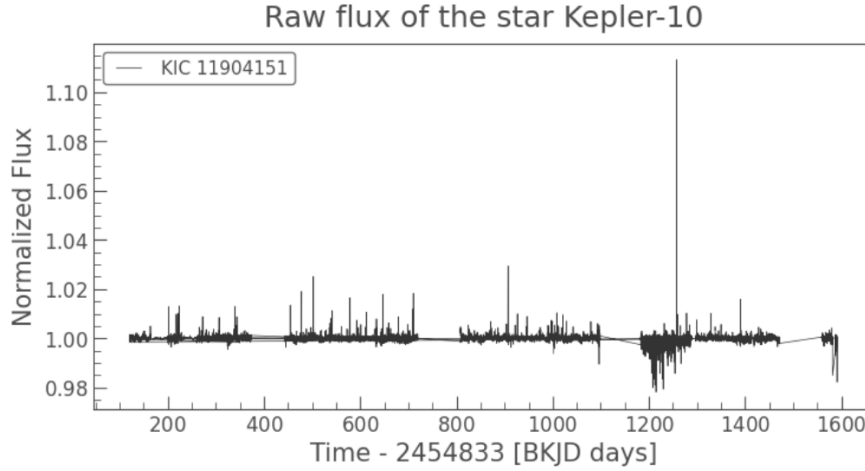


Figure 4: Raw flux of the star Kepler-10

A relationship is established between the radius of Kepler-10 b given by the experiment and that of Kepler-10 ($7.3 \cdot 10^8$) in the sense that the difference in size between the planet and its star makes this very low transit depth logical, as well as the hypotheses formulated previously. However, there is a rather significant difference between the radius of Kepler-10 b given by NASA as 9380040 meters and the result of the experiment as 7612565.4042821005 meters. Since the transit depth and the radius of the planet are linked by the following relationship $R_{planet} = R_{star}\delta$, the slight differences between the depth values from the experiment and those from NASA influenced the result for the radius of Kepler-10 b obtained in the experiment.

The experiment used one-tenth of the data stream rather than the entire stream because the calculations were too complex for Jupyter Notebook. (Figure 5)

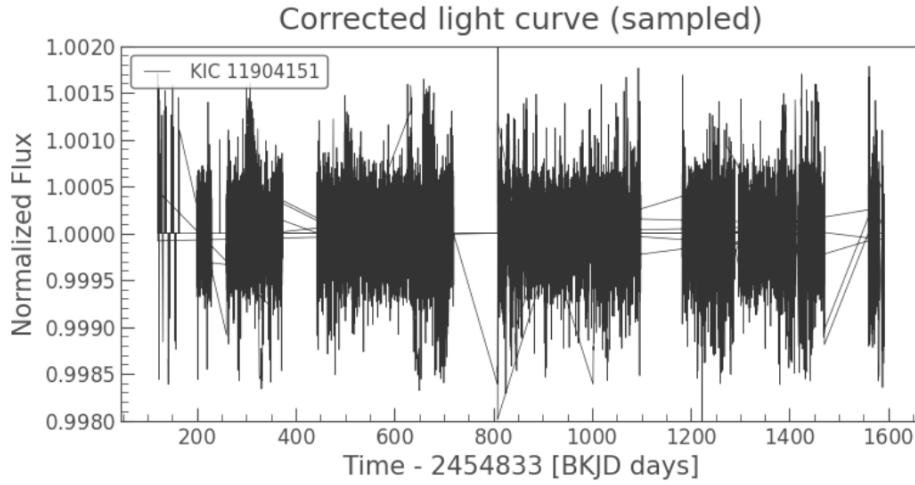


Figure 5: Corrected light curve (sampled)

Using the entire data stream would have yielded more accurate values for the transit depth and radius of Kepler-10 b. More advanced filtering and a different binning method could also have made these two values more accurate. The orbital period of Kepler-10 b could be refined by using a narrower window around the BLS maximum.

Estimating uncertainties would allow for a more rigorous comparison with the values published by the NASA. The use of a detection method other than the BLS method would enable the reliability of the results to be validated.

Conclusion

This study identified the characteristics of Kepler-10 b by analyzing data from the Kepler space mission. The values obtained are generally consistent with published data, and any differences observed may be due to noise and the analysis methods used in the experiment. These are the limitations that can be identified in the experiment. Thus, this study demonstrates the effectiveness of the BLS method for detecting and characterizing exoplanets from Kepler data.

Acknowledgements

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References

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